2(a)(i)	arrow upwards (↑) **and** labelled upthrust / U arrow downwards (↓) **and** labelled weight / W / mg arrow downwards (↓) **and** labelled tension / T *1 mark: One or two correctly labelled arrows* *2 marks: Three correctly labelled arrows*	**B2**
2(a)(ii)	$U = T + W$ **or** upthrust = tension + weight	**C1**
	$\rho Vg = T + W$ $V = [(4.00 \times 10^2) + (3.39 \times 10^4)] / (1.23 \times 9.81)$	**C1**
	$V = 2.84 \times 10^3\ \text{m}^3$	**A1**
2(a)(iii)	$m = W / g$ **or** $a = F / m$	**C1**
	$a = (4.00 \times 10^2) / [(3.39 \times 10^4) / 9.81)]$	**C1**
	$a = 0.12\ \text{m s}^{-2}$	**A1**
2(b)	there is air resistance (which increases with speed)	**B1**
	(average) resultant force is less (than weight)	**B1**
	(average) acceleration is less (than g / 9.81, so speed is less than 100 m s^{-1})	**B1**
2(c)(i)	(extension =) $2.5 \times 2.4 \times 10^{-5} = 6.0 \times 10^{-5}$ (m)	**A1**

Question	Answer	Mark
2(c)(ii)	$E_{(P)} = ½\,Fx$ **or** $E_{(P)} = ½\,kx^2$ **and** $F = kx$	**C1**
	$E_{(P)} = ½ \times 4.00 \times 10^2 \times 6.0 \times 10^{-5}$ **or** $E_{(P)} = ½ \times 6.7 \times 10^6 \times (6.0 \times 10^{-5})^2$ $E_{(P)} = 0.012\,\text{J}$	**A1**
2(c)(iii)	longer extension **or** smaller spring constant	**M1**
	elastic potential energy is greater	**A1**